

Assessment 1 – Telecom Customer & Billing Analytics

Duration: 2 Hours

Database: MySQL 8

Data Setup

```
CREATE DATABASE telecom_assessment;  
USE telecom_assessment;
```

```
CREATE TABLE customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    city VARCHAR(50),  
    mobile VARCHAR(30),  
    email VARCHAR(100)  
);
```

```
INSERT INTO customers VALUES  
(1, '  Arjun Rao ', 'Hyderabad', '98765-43210', '  ARJUN@GMAIL.COM '),  
(2, 'SARA KHAN', 'Mumbai', '+91 99887 66554', 'sara@gmail.com'),  
(3, 'Rohit Mehta ', 'Delhi', '9988 776 655', ''),  
(4, 'Neha Singh', 'Hyderabad', '9876543210', 'neha@yahoo.com'),  
(5, 'Imran Ali', 'Bangalore', '98765-AB210', NULL),  
(6, 'Priya Nair', 'Pune', '9123456789', 'PRIYA@GMAIL.COM'),  
(7, 'Kabir Shah', NULL, '9000011111', 'kabir@mail.com');
```

```
CREATE TABLE plans (  
    plan_id INT PRIMARY KEY,  
    plan_name VARCHAR(50),  
    monthly_charge DECIMAL(10,2)  
);
```

```
INSERT INTO plans VALUES  
(101, 'Basic', 399),  
(102, 'Standard', 599),
```

```
(103, 'Premium', 999),  
(104, 'Unlimited', 1499),  
(105, 'Business', 1999);
```

```
CREATE TABLE subscriptions (  
    subscription_id INT PRIMARY KEY,  
    customer_id INT,  
    plan_id INT,  
    start_date DATE,  
    status VARCHAR(20)  
);
```

```
INSERT INTO subscriptions VALUES  
(1001, 1, 103, '2026-01-01', 'Active'),  
(1002, 2, 102, '2026-01-15', 'Active'),  
(1003, 3, 101, '2026-02-01', 'Inactive'),  
(1004, 4, 104, '2026-02-10', 'Active'),  
(1005, 5, 102, '2026-03-01', 'Active'),  
(1006, 1, 105, '2026-04-01', 'Active'),  
(1007, 6, NULL, '2026-04-15', 'Pending'),  
(1008, 20, 103, '2026-05-01', 'Active');
```

```
CREATE TABLE payments (  
    payment_id INT PRIMARY KEY,  
    customer_id INT,  
    payment_date DATE,  
    amount DECIMAL(10,2)  
);
```

```
INSERT INTO payments VALUES  
(501, 1, '2026-01-05', 999),  
(502, 2, '2026-01-18', 599),  
(503, 1, '2026-02-05', 999),  
(504, 3, '2026-02-10', 399),  
(505, 4, '2026-02-15', 1499),  
(506, 2, '2026-03-18', 599),  
(507, 5, '2026-03-20', 599),  
(508, 1, '2026-04-05', 1999),  
(509, 4, '2026-04-15', 1499),  
(510, 5, '2026-05-20', 599),  
(511, 2, '2026-05-22', 599),  
(512, 1, '2026-06-05', 1999);
```

Section A – Basics & CRUD

1. Display customer name, city and email for all customers.
2. Display customers belonging to Hyderabad or Mumbai.
3. Display customers whose names contain the letter `a`.
4. Insert one new customer of your choice.
5. Update the city of customer ID 6.
6. Delete the customer inserted in Question 4.
7. Display customers ordered alphabetically by customer name.

Section B – Aggregation / GROUP BY / HAVING

8. Find the total number of payments and total amount collected.
9. Calculate total payment amount for each customer.
10. Display customers whose total payments exceed ₹2,000.
11. Find the average payment amount made by each customer.

Section C – JOINS

12. Display customer name, plan name, monthly charge and subscription status.
13. Display all customers, including customers who have no subscription.
14. Identify subscriptions having no valid customer or plan.

Section D – Data Cleansing / RegEx

15. Produce a cleaned customer output where:
 - names have leading/trailing spaces removed,
 - emails are lowercase,
 - blank emails become `NULL`,
 - mobile numbers contain only numeric characters.
 16. Using RegEx, identify mobile numbers containing alphabetic characters.
-

Assessment 2 – Airline Booking & Employee Operations

Duration: 2 Hours

Database: MySQL 8

Data Setup

```
CREATE DATABASE airline_assessment;  
USE airline_assessment;
```

Flights

```
CREATE TABLE flights (  
    flight_id INT PRIMARY KEY,  
    airline VARCHAR(50),  
    source_city VARCHAR(50),  
    destination_city VARCHAR(50),  
    ticket_price DECIMAL(10,2)  
);  
  
INSERT INTO flights VALUES  
(201, 'SkyJet', 'Hyderabad', 'Delhi', 6500),  
(202, 'AirWorld', 'Mumbai', 'Bangalore', 7200),  
(203, 'SkyJet', 'Delhi', 'Mumbai', 5800),  
(204, 'FlyHigh', 'Hyderabad', 'Dubai', 18000),  
(205, 'AirWorld', 'Bangalore', 'Delhi', 6900),  
(206, 'FlyHigh', 'Mumbai', 'Singapore', 22000),  
(207, 'SkyJet', 'Hyderabad', 'Mumbai', 5200);
```

Passengers

```
CREATE TABLE passengers (  
    passenger_id INT PRIMARY KEY,  
    passenger_name VARCHAR(100),  
    city VARCHAR(50),  
    email VARCHAR(100)  
);  
  
INSERT INTO passengers VALUES  
(1, 'Aman Verma', 'Hyderabad', ' AMAN@MAIL.COM '),  
(2, 'Sara Ali', 'Mumbai', 'sara@gmail.com'),
```

```
(3, 'Rakesh Rao', 'Delhi', ''),
(4, 'Meena Shah', 'Bangalore', 'MEENA@YAHOO.COM'),
(5, 'Farah Khan', 'Hyderabad', NULL),
(6, 'John Mathew', 'Pune', 'john@gmail.com'),
(7, 'Priya Das', NULL, 'priya@mail.com');
```

Bookings

```
CREATE TABLE bookings (
    booking_id INT PRIMARY KEY,
    passenger_id INT,
    flight_id INT,
    booking_date DATE,
    seats INT,
    status VARCHAR(20)
);
```

```
INSERT INTO bookings VALUES
(1001, 1, 201, '2026-06-01', 1, 'Confirmed'),
(1002, 2, 202, '2026-06-02', 2, 'Confirmed'),
(1003, 1, 204, '2026-06-03', 1, 'Confirmed'),
(1004, 3, 203, '2026-06-04', 1, 'Cancelled'),
(1005, 4, 205, '2026-06-05', 3, 'Confirmed'),
(1006, 5, 207, '2026-06-06', 2, 'Confirmed'),
(1007, 2, 206, '2026-06-07', 1, 'Confirmed'),
(1008, 20, 201, '2026-06-08', 1, 'Confirmed'),
(1009, 6, NULL, '2026-06-09', 2, 'Pending');
```

Employee Hierarchy

```
CREATE TABLE airline_staff (
    employee_id INT PRIMARY KEY,
    employee_name VARCHAR(100),
    manager_id INT,
    department VARCHAR(50),
    salary DECIMAL(10,2)
);
```

```
INSERT INTO airline_staff VALUES
(1, 'Raj Kumar', NULL, 'Management', 200000),
(2, 'Meera Rao', 1, 'Operations', 140000),
```

```
(3, 'Imran Khan', 1, 'Sales', 135000),  
(4, 'Aman Shah', 2, 'Operations', 90000),  
(5, 'Priya Singh', 2, 'Operations', 85000),  
(6, 'Rohit Das', 3, 'Sales', 75000),  
(7, 'Farah Ali', 3, 'Sales', 78000),  
(8, 'Vikas Rao', 4, 'Support', 60000);
```

Section A – Filtering & Manipulation

1. Display flights originating from Hyderabad.
2. Display flights priced between ₹6,000 and ₹20,000.
3. Display international-looking routes where ticket price exceeds ₹15,000.
4. Increase the ticket price of all **SkyJet** flights by 5%.
5. Display the three most expensive flights.

Section B – Aggregate / GROUP BY / HAVING

6. Find average ticket price by airline.
7. Find total number of seats booked on each flight.
8. Find airlines whose average ticket price exceeds ₹8,000.
9. Find total booking value per airline using:

```
seats × ticket_price
```

Section C – JOINS

10. Display passenger name, airline, source, destination and booking status.
11. Display all passengers including passengers who have never booked a flight.
12. Find bookings having either:
 - no valid passenger, or
 - no valid flight.

Section D – JOIN + Aggregate

13. Calculate total booking value generated by every passenger.

14. Display passengers whose confirmed booking value exceeds ₹10,000.